

Scenario 1: Image Classification Model Overfits Quickly:

You're training a CNN to classify dog breeds. You achieve 98% accuracy on training data within 5 epochs, but validation accuracy stays at 60%, and test performance is poor.

Answers:

1. Artificially increase the dataset size and diversity. Force the model to learn invariant features by applying real-time random transformations during training like Random rotations, shifting brightness etc

2. Place Dropout layers right before the final fully connected (Dense) classification layers. This randomly deactivates neurons during training, forcing the network to learn redundant representations.

3. Implement Early Stopping on validation loss.

Scenario 2: ANN Fails to Learn Non-Linear Patterns

Project: Simple digit classification using **Artificial Neural Network (ANN)**

Problem: Accuracy stuck around 50%; adding more epochs doesn't improve performance.

Answers:

1. Ensure every hidden layer uses ReLU ('relu'). Ensure your final output layer uses Softmax ('softmax') for multi-class digit classification

2. A network with zero hidden layers (input directly to output) can only learn straight lines. So add at least 1 or 2 hidden layers with 64 to 128 neurons each to give the network enough capacity to bend its decision boundaries around the digits.

Scenario 3: CNN Model Predicts Same Class for All Images

Project: Face Mask Detection using **CNN**

Problem: Model always predicts "With Mask" for every test image.

Answers:

1. Inspect your dataset count. Balance the dataset by adding more "Without Mask" images, or use class weights during training to penalize the model heavily when it misclassifies the minority class.

2. Drop the learning rate to 0.0001 and switch to an Adam optimizer.

3.Ensure the exact preprocessing pipeline (resizing, scaling to [0, 1], and color channel ordering like RGB vs BGR) is identical for training, validation, and test images.

Scenario 4: Object Detection Model Detects Objects but Bounding Boxes Are Misaligned

Project: YOLOv8 used for Tiger Detection in Forest

Problem: Bounding boxes do not fit objects properly.

Answers:

1.Ensure your coordinates are in the YOLO format: [class_id, x_center, y_center, width, height]. All four spatial values must be normalized between 0.0 and 1.0 relative to the image's total width and height. If labels are in pixel values (like Pascal VOC or COCO [xmin, ymin, xmax, ymax]), must convert them before training.

2.train with a higher resolution parameter:yolo task=detect mode=train
model=yolov8n.pt data=dataset.yaml epochs=100 imgsz=1280

Ensure that when running inference, use the exact same imgsz parameter that you used during training

3.Tighten the Intersection over Union (IoU) threshold during inference to force YOLO to merge overlapping boxes more aggressively, and raise the confidence threshold (conf).

Scenario 5: Face Mask Detection Gives High Accuracy but Performs Poorly on Real Faces

Project: Real-time Face Mask Detection using CNN

Problem: High accuracy on test set, but poor real-time performance on webcam feed.

Answers:

1.Introduce aggressive real-time data augmentations during training that mimic webcam flaws

2.Look at the cropped faces being fed into your CNN during the live webcam loop. If they look vastly different from your training images, adjust the bounding box padding in your webcam script to match the framing of dataset

3.Convert your webcam frame to RGB before passing it to the model

Scenario 6: CNN-Based Object Detector Struggles at Night

Project: YOLOv8-based object detection in CCTV

Problem: Model doesn't detect objects in dark/night images.

Answers:

1. Drop the brightness and contrast parameters significantly in YOLO training configurations to mimic poor lighting.

2. Using CLAHE balances local contrast natively, bringing out hidden edge details of objects moving through dark shadows without blowing out overexposed areas like streetlights.

3. Slicing Aided Hyper Inference (SAHI) slices the massive CCTV frame into smaller overlapping patches, resizing each patch to the network's optimal resolution (imgsz=640) before performing detection.